

AGING

Circadian alignment of early onset caloric restriction promotes longevity in male C57BL/6J mice

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Caloric restriction (CR) prolongs life span, yet the mechanisms by which it does so remain poorly understood. Under CR, mice self-impose chronic cycles of 2-hour feeding and 22-hour fasting, raising the question of if it is calories, fasting, or time of day that is the cause of this increased life span. We show here that 30% CR was sufficient to extend the life span by 10%; however, a daily fasting interval and circadian alignment of feeding acted together to extend life span by 35% in male C57BL/6J mice. These effects were independent of body weight. Aging induced widespread increases in gene expression associated with inflammation and decreases in the expression of genes encoding components of metabolic pathways in liver from ad libitum-fed mice. CR at night ameliorated these aging-related changes. Our results show that circadian interventions promote longevity and provide a perspective to further explore mechanisms of aging.

Caloric restriction (CR) without malnutrition or starvation, which is achieved by reducing ~30% of daily food intake, is the most effective nonpharmacological intervention that improves life span in model organisms (1), but the underlying mechanisms remain unclear (2–6). Classical CR protocols in mice lead to a temporal restriction of food intake with a long (>22 h) fasting interval because mice consume the food as soon it becomes available (7–9). Timed food administration is a potent signal that entrains circadian clocks in peripheral tissues such as liver (10–12). Thus, in addition to reducing daily energy intake, CR resets complex circadian programs of gene expression in tissues throughout the body (13–15). Although decreased energy intake is commonly thought to be the critical factor that extends life span, it is possible that the timing of food intake is also a key component. The changes caused by time-restricted feeding can have profound effects on physiology (16). For example, mice (which are nocturnal) fed a high-fat diet only during the day gained significantly more weight than mice fed the same diet only during the night (17). Also, mice fed a high-fat diet restricted to an 8-hour window during the night were protected against diet-induced obesity, hepatic steatosis, hyperinsulinemia, and inflammation compared with mice fed ad libitum (AL) (18, 19). Thus, temporally restricted feeding at night, which is the normal active and feeding time of day for mice, is beneficial.

Although the timing of food intake can have an impact on health, it remains unclear whether the timing and frequency of feeding also affect life span in mice (16, 20, 21). Food consumption triggers behavioral and metabolic changes in mammals that have profound impacts on health status (22). We studied the contributions of feeding time and fasting under CR and compared behavioral, metabolic, and molecular outcomes throughout the life span. We tested five different CR protocols and an AL control group using automated feeders (7). After 6 weeks of baseline AL food access, C57BL/6J male mice were subjected to 30% CR. Mice were fed nine to ten 300-mg food pellets containing 9.72 to 10.8 kcal every 24 h starting at the beginning of the day (CR-day) or night (CR-night), similar to classical protocols in which mice consumed their food within 2 h as one meal (7). To prevent the 2-h binge-eating pattern and to reduce the fasting interval to ~12 h, two additional CR groups of mice were fed a single 300-mg pellet (1.08 kcal) delivered every 90 min to distribute the food access over a 12-h window either during the day (CR-day-12h) or during the night (CR-night-12h). A fifth CR group of mice was fed a single 300-mg pellet every 160 min continuously spread out over 24 h (CR-spread) to abolish the rhythmic pattern of food intake and to prevent any fasting intervals (Fig. 1A).

Behavioral and body weight dynamics with age

To select the diet for the longevity studies, we first compared standard laboratory chow (Teklad Global 2018) with two different precision food pellets with similar caloric content but different compositions: a grain-based diet that we used previously (F0170) (7) and a purified diet (F0075) (fig. S1A). We found that mice fed the purified diet showed body weight gain similar to the standard laboratory chow;

however, the mice fed grain-based pellets gained significantly more weight (fig. S1B). Because the composition of grain-based diets is known to vary by batch and by season of the year (23) and because longevity experiments require at least 4 years, we chose the purified diet that could be completely defined and maintained over the entire duration of the life-span experiments and did not cause excessive weight gain compared with standard laboratory chow. We used an automated feeding system (7) and monitored feeding and wheel-running activity of individually housed mice continuously throughout their life span. This allowed us to measure behavioral and metabolic changes in mice under all six feeding conditions as they aged.

In agreement with previous studies (24), mice under unrestricted feeding (AL) gradually increased their body weight until 20 months of age, after which they showed an age-related decline (Fig. 1B and fig. S2). All CR groups maintained lower body weights throughout their life span, consistent with lower food intake (Fig. 1B and fig. S3). We previously showed that CR-day mice gained more weight than CR-night mice with the grain-based diet (7), but this effect was not reproduced with the purified diet (fig. S2), perhaps because of the difference in fat source in the two diets (fig. S1A). Long-term recordings of feeding events showed that mice adjusted their feeding patterns to match the externally controlled availability of food (including daytime feeding and 24-h spread-out feeding). These feeding patterns were consistently maintained throughout their life span (Fig. 1, C and E; fig. S4A; and data S1). Mice in the AL group normally consumed ~75% of their food at night and maintained this pattern of food consumption throughout their life span, with a gradual increase in food consumption with age after the first year (Fig. 1F and fig. S4A). Mice in the CR-night-2h and CR-day-2h groups with 24-h access to 30% CR (relative to AL controls for the first 200 days of study) rapidly consumed their daily allotment within 2 h, as previously described (Fig. 1, C and E) (7), and this 2-hour intake pattern was maintained throughout their life span (fig. S4A). Although AL mice increased their food consumption after 1 year of age, the amount of food was not increased for the CR groups (fig. S3), so the CR increased from 30 to ~40% compared with AL at later ages. Similarly, animals exposed to CR with food access spread over 12 or 24 h also adapted to the imposed meal pattern by eating each pellet as soon as it became available, which was every 90 min (CR-day-12h and CR-night-12h) or every 160 min (CR-spread) (Fig. 1, C and E, and figs. S3 and S4A). When examining the median phase of feeding, which is the time at which mice ate 50% of their daily allotment, we observed that the phase for classic CRs

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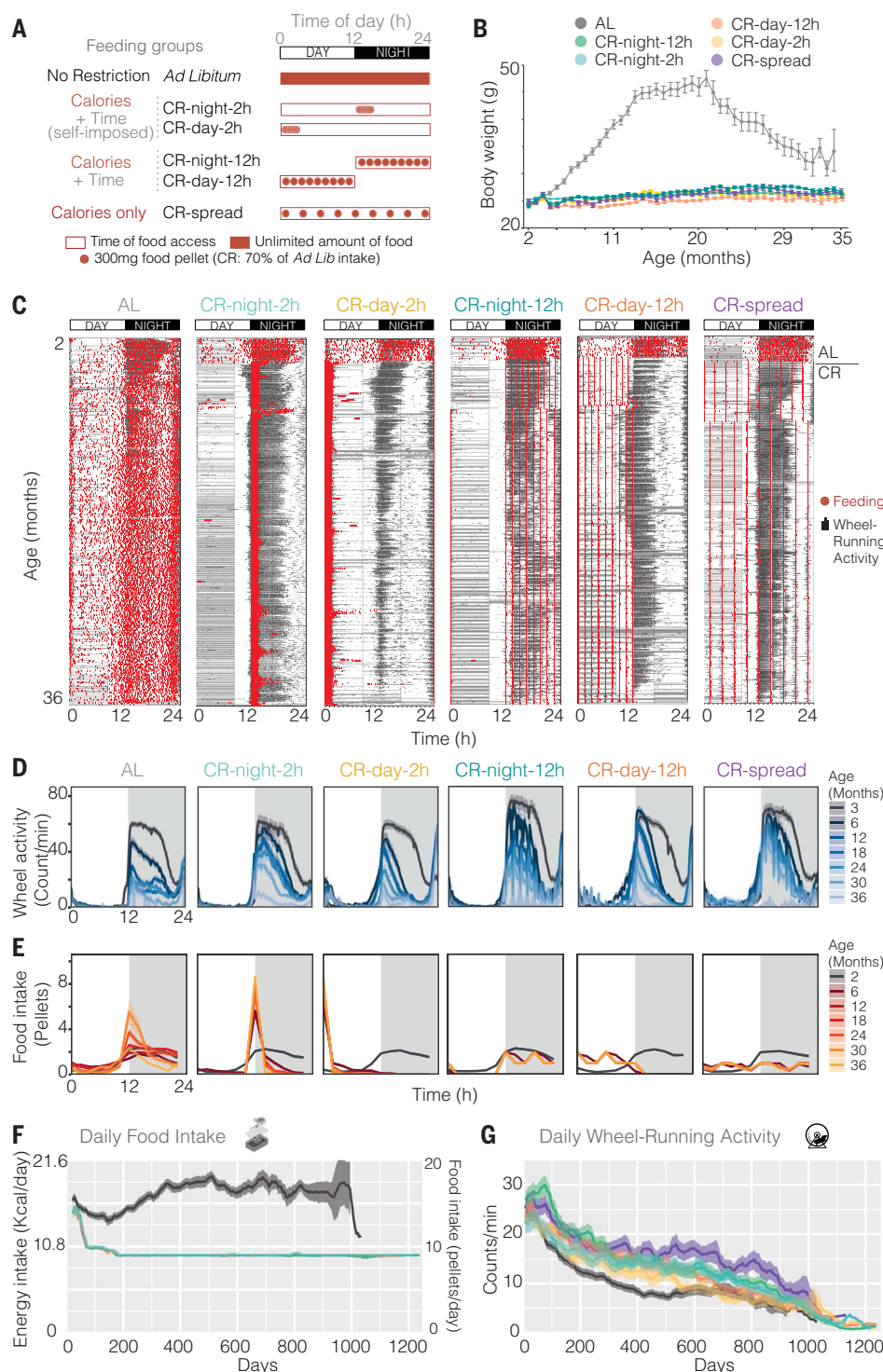


Fig. 1. Effects of CR on body weight, circadian behavior, and feeding in C57BL/6J male mice. (A) Experimental design showing feeding conditions for each of the six groups. (B) Average body weights ($\pm$ SE) of mice in all six groups ($n = 43$ for the AL group and $n = 36$ for each of the CR groups) taken every 3 weeks throughout the experiment. (C) Examples of double-plotted actograms from each experimental group overlaying wheel-running (black histograms) and feeding (red dots) behaviors. All mice were on AL feeding for the first 6 weeks (period above the line on the right of the actograms) before the CR began. (D and E) Twenty-four-hour profile of wheel-running activity (D) and food intake (E) at different ages (averaged over 21 days, $n = 36$ to 43 mice) for each group. (F) Energy intake per day (left axis) and number of food pellets per day (right axis) for each group throughout the experiment. For the AL group, the dark line is average and the gray shading is SE. All CR groups were limited to 70% of AL consumption for the first 200 days of age and were not adjusted after 200 days, so no variation is observed. (G) Daily wheel-running activity (average counts/min over 24 hours $\pm$ SE) throughout the experiment.

was 1 h after the food onset [Zeitgeber time (ZT) 1 h for CR-day-2h and ZT 13 h for the CR-night-2h]. For the CR-spread group, the phase was ZT12 because the mice ate equal amounts during the day and night (fig. S4A). The feeding pattern was also consistent with daily changes in body weight (fig. S4B). With the exception of the CR-spread group, in which the food was equally distributed throughout

the day, body weight changed throughout 24 h with a significant increase during the feeding time (fig. S4B). This finding was more pronounced in classic CR protocols, with the highest body weight gain of 3 g occurring between ZT0 and ZT4 in CR-day-2h mice and between ZT12 and ZT16 in CR-night-2h mice. These body weight gains are consistent with the observation that mice eat their

entire allotment (2.7 to 3 g) as one single meal within 2 h.

All groups maintained a normal nocturnal locomotor activity pattern for life, with the exception of the day-fed mice, which tended to have more daytime activity (Fig. 1, C and D). Overall, these long-term recordings showed that when food was restricted to the daytime, mice interrupted their “rest phase” to eat but

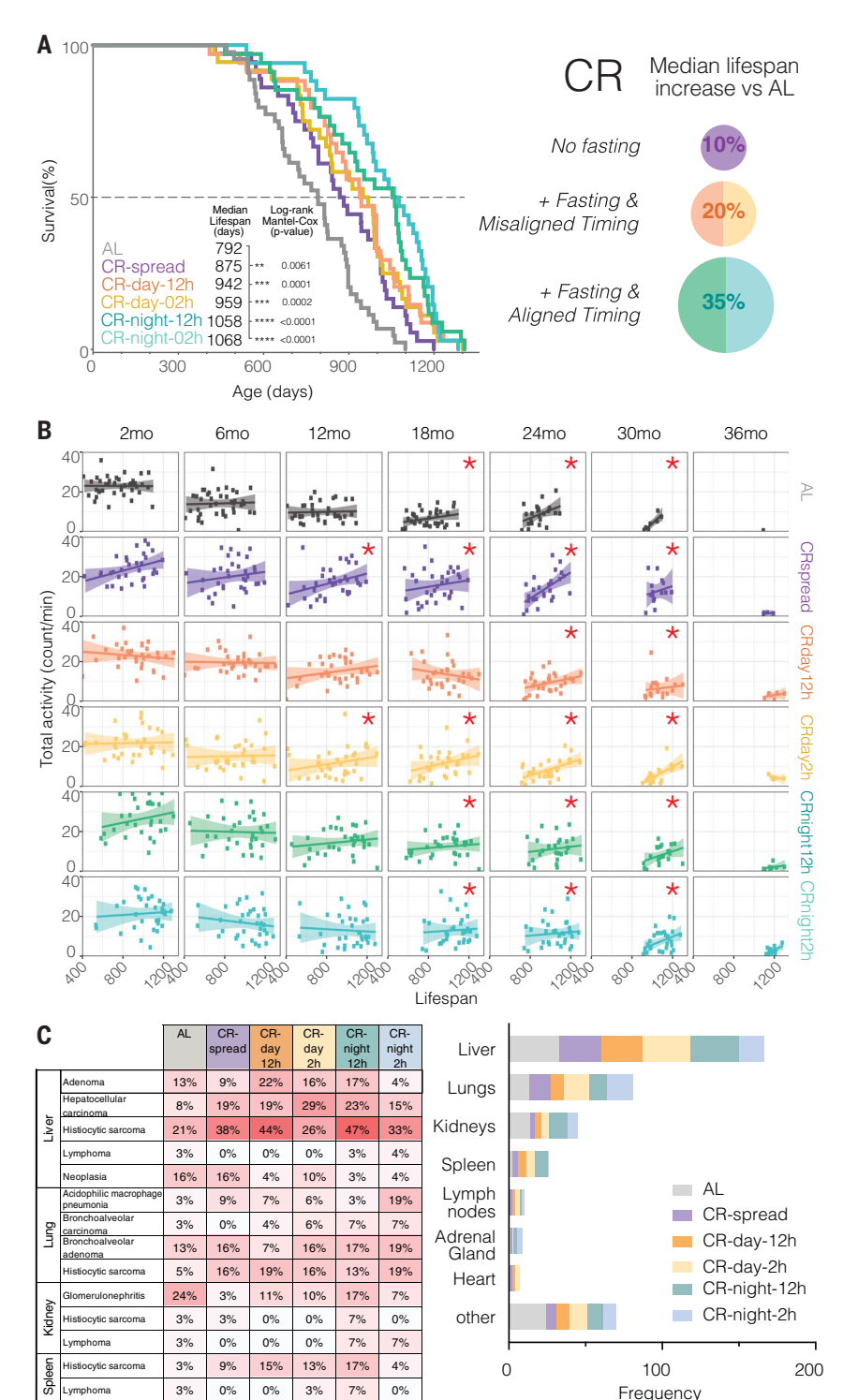


Fig. 2. Extent of CR-mediated increases in longevity depend on feeding time. (A) Survival curves for each group ($n = 43$ for AL and $n = 36$ for each of the CR groups) are shown in the left panel and the median life span (days) is shown in the inset. Right panel summarizes the results, showing the increase in life span from timed feeding with the largest increase when food is restricted to night. (B) Correlation plots comparing life span (days) for each mouse with its daily averaged total activity (counts/min) at different ages. Increased activity significantly correlates with longer life span in old, but not young, mice in all groups (see asterisks, Spearman correlation). (C) Necropsy followed by histopathology results showing pathologies and diseases (left) and tissues mostly affected (right) at the time of death for each feeding condition.

maintained most activity during the nighttime (Fig. 1, C and D). Therefore, feeding and locomotor activity are misaligned with daytime feeding in these animals throughout their life span, which would be expected to lead to adverse metabolic consequences (25, 26). Activity of the mice declined as they aged (27), with AL mice having the lowest activity levels compared with the CR groups between 6 and 18 months of age [Fig. 1G and fig. S5; two-way analysis of variance (ANOVA); age, $P < 0.0001$; feeding, $P < 0.0001$; interaction nonsignificant (NS)].

Life-span extension by CR depends on feeding time

We investigated the contribution of calories, feeding time, and fasting period on longevity. CR was sufficient to extend the median life span in male mice, but the range of this extension depended on when the food was consumed (Fig. 2A). The percentage of life-span extension varied across conditions. Consistent with other reports, AL mice had a median life span of 792 days (24, 28). CR-fed mice lived 10 to 35% longer than AL mice depending on the CR group. The CR-spread group, which had a 30% reduction in calories but with feeding spread throughout the day-night cycle, had a median life span of 875 days, which is 10.5% longer than that of AL mice, demonstrating that CR alone without time restriction or fasting is sufficient to extend longevity. The CR-day-12h and CR-day-2h groups had median life spans of 942 and 959 days, respectively, which are 18.9 and 21.1% longer than the life spans of AL mice, respectively. Thus, in addition to the reduction in calories, a minimum of 12 hours of fasting induces its own benefits on longevity. There were no significant differences in life span when day-fed mice fasted for ~22 hours versus 12 hours, indicating that 12 hours of fasting is sufficient. CR-night fed mice outlived both CR-day groups: The CR-night-12h and CR-night-2h mice had median life spans of 1058 and 1068 days, respectively, which are 33.6 and 34.8% longer than the life spans of AL mice, respectively. Again, there was no additional benefit of ~22 hours of fasting compared with 12 hours of fasting in these groups fed at night, indicating that 12 hours of fasting is sufficient for prolonging life span. There was a significant extension of life span by CR-night-2h mice (34.8% extension) over CR-day-2h mice (21.1% extension) (log-rank Mantel-Cox, $P < 0.05$), which differed in the relative phase of food consumption by the mice. It is possible that sleep disruption due to misaligned feeding could contribute to the difference in life span. Further studies are required to determine whether sleep is affected and, if so, if potential sleep disruptions can contribute to the differences in life span observed between CR-day

versus CR-night animals. However, it has recently been shown that sleep homeostasis is maintained in mice under a restricted feeding schedule in which food is only available for 4 h in the middle of the “sleep” phase (ZT4 to ZT8) (29). In all five of the CR groups in this study, the mice consumed exactly the same number of daily calories throughout their life span (figs. S3 and S4; two-way ANOVA; age, $P < 0.0001$; feeding, $P < 0.0001$; interaction, $P < 0.0001$), yet the pattern and circadian phase of feeding had major effects on life span. A >12-hour fasting interval combined with nocturnal (normal) feeding yielded the greatest benefits on life span. Thus, calories are processed differently depending on when they are consumed, and anti-aging interventions such as CR can be optimized by timing them to a specific time of day. Maximum life span, estimated as the 10% longest-lived mice in each group, was significantly longer in all of the CR groups compared with AL except for the CR-spread group (exact Fisher’s test, $P < 0.05$) (data S1). Among the CR groups, only CR-night had a significant increase in maximum life span compared with CR-spread (exact Fisher’s test, $P = 0.0256$ CR-night versus CR-spread). This suggests that feeding and fasting cycles that are in sync with internal circadian clocks (CR-night-2h) extend both the median and maximum life span, which is indicative of delaying the aging process as opposed to delaying the onset of a single disease.

Necropsy followed by histopathology revealed that all groups had similar diseases at death, but in the CR groups, these diseases occurred at older ages (coinciding with longer life spans) as previously reported (8). Neoplasias were the most frequent pathology in all groups, with histiocytic sarcomas being the major cause of death, followed by hepatocellular carcinoma (Fig. 2C and data S1). Histopathology analysis of the target tissues also revealed the highest incidence of lesions in the liver (Fig. 2C).

Age-related decline in activity predicts lower survival in mice

To evaluate whether behavioral or metabolic parameters correlated with longer life spans, we compared feeding (total daily intake), body weight, and wheel-running activity (total daily activity and percentage nighttime activity) with life span. We found no correlation of life span with food intake or body weight at any age (figs. S6 and S7). However, in all feeding conditions, daily locomotor activity level positively correlated with longer life span after 18 months of age (Fig. 2B). Additionally, higher activity levels during the normal circadian phase (at night) after 24 months of age also positively correlated with longer life span (figs. S8 and S9). Because voluntary wheel-running activity does not affect life span in mice (30),

our results suggest that the level of wheel-running activity after 18 months of age could be a biomarker for health span. Thus, the activity level of mice at older ages (>18 months) can be used as a predictor of longer life span.

CR promotes widespread metabolic benefits

Body composition analysis at 12 and 20 months of age showed that although fat mass was significantly higher in the AL group versus the CR groups, there were no differences in fat mass among the CR groups (fig. S10). We assessed metabolic markers from plasma at 6 and 19 months of age. Insulin levels increased with age under AL, and such increases were attenuated by all CR groups, which maintained low insulin levels at both ages (fig. S11). In young mice, CR groups had similar insulin levels yet lower glucose levels in plasma compared with the AL group (figs. S10 and S11), suggesting that improved insulin sensitivity was associated with lower food intake. As the mice aged, similar levels of circulating glucose were found in all feeding groups even though AL-fed mice had higher insulin levels than CR groups, suggesting that CR generally protects against age-related insulin resistance. Similar to what was seen with insulin, there was a significant age-related increase in leptin under AL that did not occur in the CR groups. Although all CR groups maintained lower leptin levels than the AL group at 19 months of age, only the CR-night groups had leptin levels that were significantly lower at younger ages, suggesting that alignment of feeding may play a role in regulating satiety at both ages. Glucagon-like peptide 1 (GLP-1) is an intestinal hormone that is secreted after meals and decreases blood sugar levels by promoting insulin secretion and suppressing glucagon release (31). We found that although the levels of GLP-1 did not change with aging, the longest-lived (CR-night) groups had significantly lower levels at older ages compared with AL (fig. S11). This indicates that CR-night mice may have had an improved sensitivity to GLP-1 in regulating insulin secretion and glucose levels. Overall, the longest-lived CR groups had improved hormonal profiles, insulin sensitivity, and glucose homeostasis as they aged.

Gene expression changes with aging and CR

To determine the effects of aging and feeding at the molecular level, we assessed circadian gene expression patterns using RNA sequencing (RNA-seq) in mouse liver in all six feeding conditions at two ages: young mice at 6 months and old mice at 19 months. We chose to profile the liver because it is a major metabolic target of the circadian clock system (32) and because it had the highest incidence of age-related lesions among target tissues (Fig. 2C). We chose 6 months of age to assess fully mature adult mice and 19 months of age to assess aged

mice before their obvious decline in body weight (Fig. 1B) and before there was >10% mortality in AL mice (Fig. 2A). In each group, we profiled the liver at 12 time points every 4 hours for 48 hours across two circadian cycles in mice transferred to constant darkness to assess circadian rather than diurnal cycles (two biological replicates $\times$ 12 time points = 24 liver samples per feeding condition). We chose to perform these gene expression experiments in constant darkness to determine whether a rhythmic gene profile is circadian. Because liver-cycling gene expression patterns are strongly driven by feeding cycles (33), the absence of light:dark (LD) cycles would be expected to yield results for rhythmic gene expression in the liver similar to those seen in LD cycles reported previously (11, 33). Unbiased principal component analysis showed that (i) samples from young and old groups clustered separately under AL, (ii) young groups under CR clustered together and separately from AL, and (iii) old groups under CR clustered together between aged-AL and young-CR groups (Fig. 3A). This suggests that at the molecular level, the aging process is different between mice fed AL or CR, and aged-CR mice maintained a liver gene expression pattern more similar to that of the young animals.

To address the overall impact of aging at the molecular level, we performed differential gene expression analysis between young and old mice (data S2 and S3). A total of 2599 genes (18.6% of expressed genes) were differentially expressed with aging under AL feeding (Fig. 3B and fig. S12). Of these, 2031 genes were up-regulated and 568 were down-regulated. Gene ontology (GO) analysis revealed that the up-regulated genes were highly significantly related to immune system processes and inflammation (Fig. 3C), whereas the down-regulated genes were related to metabolism (Fig. 3D and data S4). This is consistent with previous reports showing that increased inflammation and senescence are hallmarks of aging (3) and recent work on glycine-serine-threonine metabolism in longevity (34). Among the up-regulated genes were *Cd36* ($\log_2FC = 3.8$, adjusted $P = 2.95 \times 10^{-92}$), which is a multifunctional glycoprotein that acts as a receptor for ligands such as thrombospondin, fibronectin, amyloid beta, oxidized low-density lipoprotein, and long-chain fatty acids, among others (35), and the peroxisome proliferator-activated receptor gamma (*Pparg*, $\log_2FC = 1.8$, adjusted $P = 1.5 \times 10^{-28}$), a transcription factor also associated with immunosenescence during aging (Fig. 3C) (36, 37). Other genes of note were *Adora1*, *Serpine1*, *Themis*, 10 Toll-like receptor (*Tlr*) genes, *Spon2*, and *Zcchc11* (data S2 and S3). Among the down-regulated genes, there was an enrichment in key enzymes of amino acid (*Gnmt* and *Agxt*) and cholesterol metabolism such as HMG-CoA reductase

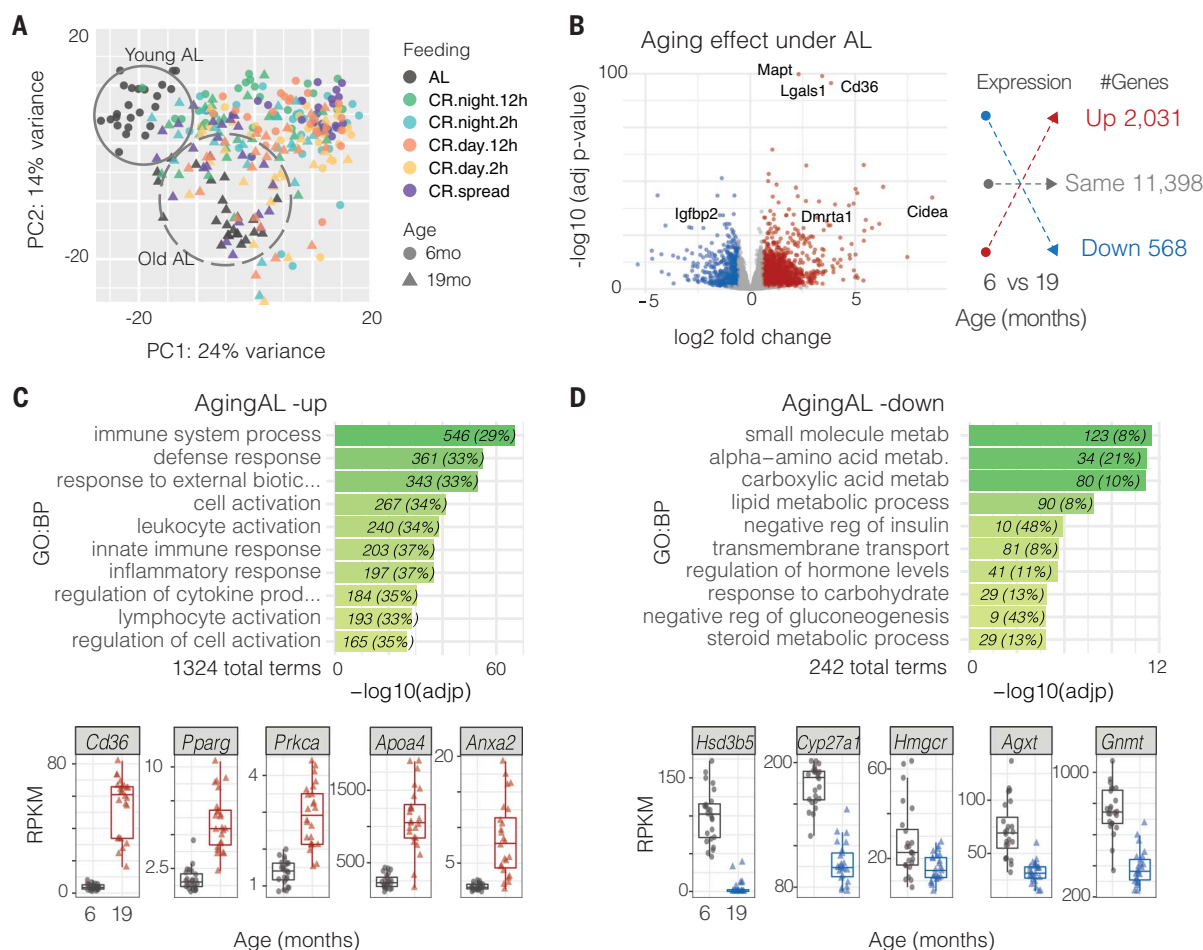


Fig. 3. Gene expression signatures in liver change during aging in AL mice.

(A) Principal component analysis of gene expression from liver mRNA-seq. mRNA-seq data are from 48 mice for each feeding condition ($n = 24$ from 6 months of age, $n = 24$ from 19 months of age), with livers collected every 4 hours over 48 hours while mice were in constant dark. Circles indicating young AL mice (solid line) are clustered together, and triangles indicating aged AL mice (dashed line) are in a distinct cluster. Liver gene expression data among CR groups cluster together independently of age. (B) Volcano plot showing

differential gene expression in young versus aged AL mice. Red denotes genes for which expression is significantly increased in aged AL mice; blue denotes genes that are significantly decreased in old AL mice. The number of mRNAs in each category are shown in the right panel. (C) GO terms of genes that are increased in aged AL mice (top) and examples of gene expression (bottom). Gray indicates young AL, and red indicates aged AL. (D) GO terms of genes that are decreased in aged AL mice (top) and examples of gene expression (bottom). Gray indicates young AL, and blue indicates aged AL.

(*Hmgcr*), which is implicated in liver disease and hepatocellular carcinoma (38–41) (Fig. 3D). Other metabolic genes of note were *Got*, *Lepr*, *Lpin1*, *Pfkfb3*, *Scap*, *Hsd17b2*, *Hsd3b5*, as well as 14 cytochrome P450 (*Cyp*) genes and 28 solute carrier (*Slc*) genes (data S2 and S3). Overall, there was an increase in the expression of inflammatory genes and a decrease in the expression of metabolic genes.

CR alone rescues most age-related changes observed under AL

To determine the overall effect of CR, we analyzed which genes underwent age-related changes in any condition and identified those with expression levels that remained protected under CR versus AL. Across all feeding conditions, there were 4077 genes with expression that was up- or down-regulated with aging

(Fig. 4A and data S2), indicating that 29% of the liver transcriptome is susceptible to aging-related changes under any condition tested. AL-fed mice had the highest percentage of genes that changed with aging (18%), whereas all of the CR groups had lower overall changes in gene expression with age (Fig. 4A). The CR-night-2h group had the smallest overall change in gene expression with age (4%) (Fig. 4A). Age-related fold change of individual genes in the CR-spread and CR-day groups were partially decreased compared with the fold changes seen in AL, whereas CR-night-2h strongly attenuated these age-related changes. Figure 4B shows these results as correlation plots of age-related fold change compared with AL for all CR groups. If there were no rescue by CR, the data points would fall on the unity line (slope = 1), whereas with complete rescue by CR, the

data points would fall on the horizontal line (slope = 0). As seen in Fig. 4B, the regression line for the CR-night-2h group was almost completely flat, demonstrating that CR-night-2h strongly reduced age-dependent changes in gene expression. Approximately 50% of age-related changes in gene expression under AL (1233 genes) were restored in every CR condition (Fig. 4C and data S2) by rescuing 44% of up-regulated genes (inflammation and immune function) and 60% of down-regulated genes (metabolic pathways). This result indicates that CR alone prevents most of the age-related changes observed in the control AL condition.

GO analysis showed that the genes protected by CR were also associated with immune function, inflammation, and metabolism (Fig. 4C and data S4) (34). Among these genes were those

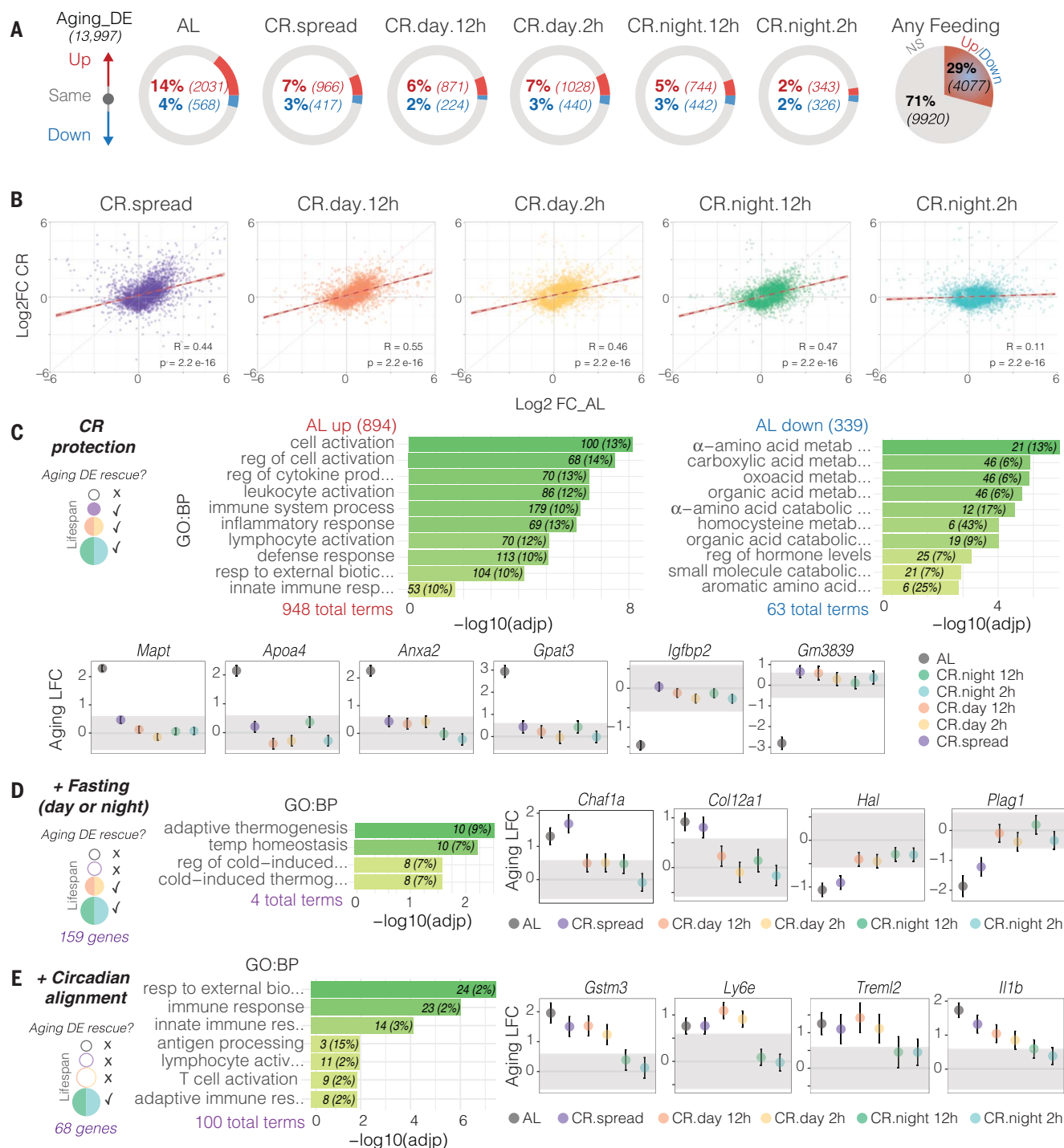


Fig. 4. CR ameliorates age-related changes in liver gene expression observed under AL conditions. (A) Schematic comparison of differential gene expression between young and old mice in the six feeding conditions. Circles show the percentage of genes that are unchanged between young and aged mice (gray), increased in aged mice (red), and decreased in aged mice (blue). Pie chart on the right shows the percentage of genes susceptible for age-related changes in any feeding condition. (B) Spearman correlation plots comparing changes in gene expression between the aging DE genes between AL and CR groups (aging DE genes are defined here as the 4077 genes that change with age in any of the six feeding conditions tested).

(C) Schematic representation of genes that are protected from age-related changes in every CR group (left) with GO terms of those significantly up-regulated (middle) or down-regulated (right) in aged AL mice. Represented are 10 nonredundant of the top 25 most significant enriched terms. Examples of age-related fold changes (log₂FC ± SE) in gene expression are shown below for all feeding conditions. Gray-shaded areas indicate FC < 1.5 considered as not significant change. (D and E) Schematic representation, GO, and representative genes that maintain similar levels between young and old ages due to fasting (day or night) (D) and circadian alignment of feeding and fasting cycles (E).

encoding for microtubule-associated protein tau (*Mapt*) and apolipoprotein A-IV (*ApoA4*), the expression of which was up-regulated in old-AL mice but maintained at lower (young) expression levels in all CR groups (Fig. 4C). These genes have been linked with aging and neurodegeneration in Alzheimer's disease (3, 42). A similar aged-related up-regulation occurred in the liver and was reduced by CR. Of the metabolic genes that declined with age under AL, such as insulin-like growth factor-binding protein 2 (*Igfbp2*), CR strongly attenuated changes in expression in all five CR groups.

Fasting-related genes

To evaluate gene expression changes caused by fasting, we compared genes that were differentially expressed only in AL and CR-spread but remained constant in all the other four CR groups with some degree of fasting. We found 159 genes of this type that could be potential candidates accounting for the change from 10 to 20% life-span extension (Fig. 4D and data S2). Among these were collagen type XII alpha 1 chain (*Col12a1*), which is associated with cancer (43) and is elevated in liver fibrosis, and pleomorphic adenoma gene 1 (*Plag1*), an oncogene associated with hepatoblastoma and age-related decrease in skeletal muscle (44, 45). Other genes included chromatin assembly factor 1 subunit A (*Chaf1a*) (46) and histidine ammonia-lyase (*Hal*) involved in amino acid metabolism. GO analysis revealed that these fasting-related genes were also enriched for thermogenesis pathways.

Time-related genes

To evaluate the beneficial effect of feeding time, we searched for genes that maintained similar levels at young and old ages only in the CR-night fed groups with the longest life spans but were differentially expressed in the AL, CR-spread, and CR-day fed groups. We found 68 genes that were specifically protected in the CR-night groups (Fig. 4E and data S2). GO analysis revealed specific subsets of genes involved in the immune system and inflammation, such as glutathione S-transferase mu 3 (*Gstm3*), which protects against oxidative stress; lymphocyte antigen 6 family member E (*Ly6e*), which regulates T-cell proliferation, differentiation, and activation; triggering receptor expressed on myeloid cells-like 2 (*Trem2*); and proinflammatory cytokines such as interleukin-1 β (*Il1b*) that are increased in the elderly (3). Thus, circadian alignment of feeding time adds another level of protection of immune function and age-related inflammation beyond that seen with CR and fasting.

Circadian cycling of gene expression with aging

Robust circadian rhythms are at the core of a healthy physiology, and rhythm amplitude decreases in response to infectious diseases

and aging (22, 47, 48). We investigated how feeding conditions, timing, and CR influenced circadian cycling of gene expression in young and aged mice. To search for circadian cycling genes, we performed RNA-seq from liver samples collected every 4 hours for 48 hours (data S4 and S5). We used very strict criteria to identify circadian cycling genes by selecting only those genes that were significant in three out of three different algorithms (JTK_CYCLE, ARSER, and RAIN) with a P value < 0.05 and a false discovery rate (FDR) < 5% (FDR < 0.05). We found 1718 rhythmic genes in young AL mice and 1507 rhythmic genes in old AL mice (see Fig. 5A, heatmaps of cycling genes, and data S5). The overlap of these two genes sets was 694 genes. Therefore, circadian cycling genes were both lost and gained with age (Fig. 5A). Figure 5B illustrates six cycling genes in young and old mice. Of the four circadian clock genes, *Arntl*, *Nr1d1*, *Per1*, and *Per2*, three had a lower amplitude in old mice. Two metabolic pathway genes, *Gys2* and *Pck1*, also had lower amplitude with age consistent with the overall decline in average gene expression in metabolic pathways (Fig. 3, B and D; see fig. S13 for example circadian profiles of pro-aging and pro-longevity genes). We compared the circadian phase and amplitude (fold change between trough and peak on the first and second circadian cycle) for the 694 shared cycling genes in young and old mice (Fig. 5A). There was no change in the phases of cycling genes with age; however, the amplitude of these cycling genes was lower, as seen by the deviation of the regression line from unity (slope = 0.5941 ± 0.009 , $P < 0.0001$) (Fig. 5C). GO analysis of these cycling gene sets showed enrichment for metabolism, cell communication, signaling, and circadian rhythms (Fig. 5, D and E, and data S3), which is consistent with the decline in average gene expression in metabolic pathways (Fig. 3D) and with analysis of circadian gene targets using chromatin immunoprecipitation sequencing, in which the GO category metabolism was highly significant (32).

Effects of day versus night CR on circadian gene expression

Because the median life span of mice in the CR-night-2h and CR-day-2h groups was significantly different (1068 versus 959 days, respectively, and log-rank Mantel-Cox, $P < 0.05$), we focused on differential gene expression in these two CR groups. One essential difference in these two groups is the phase of food consumption relative to the LD cycle and relative to the circadian phase of the mice as assessed by their locomotor activity rhythms. We selected the sum total genes that showed circadian cycling in at least one of these four groups: CR-day-2h and CR-night-2h from young and old mice, which led to a total of 1491 cycling

genes. Figure 6A shows cycling genes in these four groups as a heatmap in which each line represents the color-coded levels of expression of one gene (in each row) across time points (columns). There was a slight reduction in cycling genes in young CR-day-2h mice, but in old CR-day-2h mice, there were only seven cycling genes. Circadian gene expression profiles of the six genes shown in Fig. 5C showed that the day-fed groups at 19 months of age have either opposite phases (*Arntl*, *Nr1d1*, *Gys2*, *Per2*, and *Pck1*) or disrupted circadian profiles (*Per1*) (see Fig. 6B and fig. S13 for example circadian profiles of pro-aging and pro-longevity genes). We compared the fold-change amplitude of the 1491 genes in young and old mice. At both 6 and 19 months of age, the CR-night-2h groups had a significantly higher amplitude compared with the CR-day-2h groups, as seen in the fold-change frequency histograms and correlation plots (Fig. 6C). Thus, CR-night-2h feeding enhanced circadian amplitude relative to that of CR-day-2h.

To assess the phase of entrainment in these four CR groups, we compared the phases of cycling genes relative to those of AL mice at 6 and 19 months of age. Figure 6D shows correlation plots of genes that overlapped with AL mice in either the CR-day-2h or CR-night-2h groups at 6 and 19 months of age. The CR-night-2h mice shared many more cycling genes with AL mice than did the CR-day-2h mice at both ages. In addition, the phases of gene expression of CR-night-2h mice of both ages were correlated with the phases of the genes from the respective AL age group, with the data points falling on the unity line. As expected, the CR-day-2h mice had opposite or divergent phases relative to those of AL mice. For the old CR-day-2h mice, only seven genes were scored as cycling and only five overlapped with those of AL mice. Thus, in CR-day-2h mice, there was a paucity of cycling genes and this declined precipitously with age. A similar reduction was seen in fold-change gene expression in CR-spread mice compared with CR-night-2h mice at 6 and 19 months of age (fig. S14). The CR-night-2h groups showed robust circadian cycling, even in the old mice, suggesting that this intervention is very effective in rescuing circadian cycling of gene expression relative to CR-day-2h. A proviso to this cycling analysis is that these cycling algorithms, JTK_CYCLE and ARSER, are biased toward sinusoidal waveforms, and although RAIN can search for spiky or sawtooth waveforms, our requirement that a cycling gene is significant (FDR < 0.05) in all three algorithms results in only genes with sinusoidal waveforms passing this threshold. Daytime feeding and 2-hour feeding bouts both modified the gene expression time series waveforms so that they were less sinusoidal, and this contributed to the lower number of genes

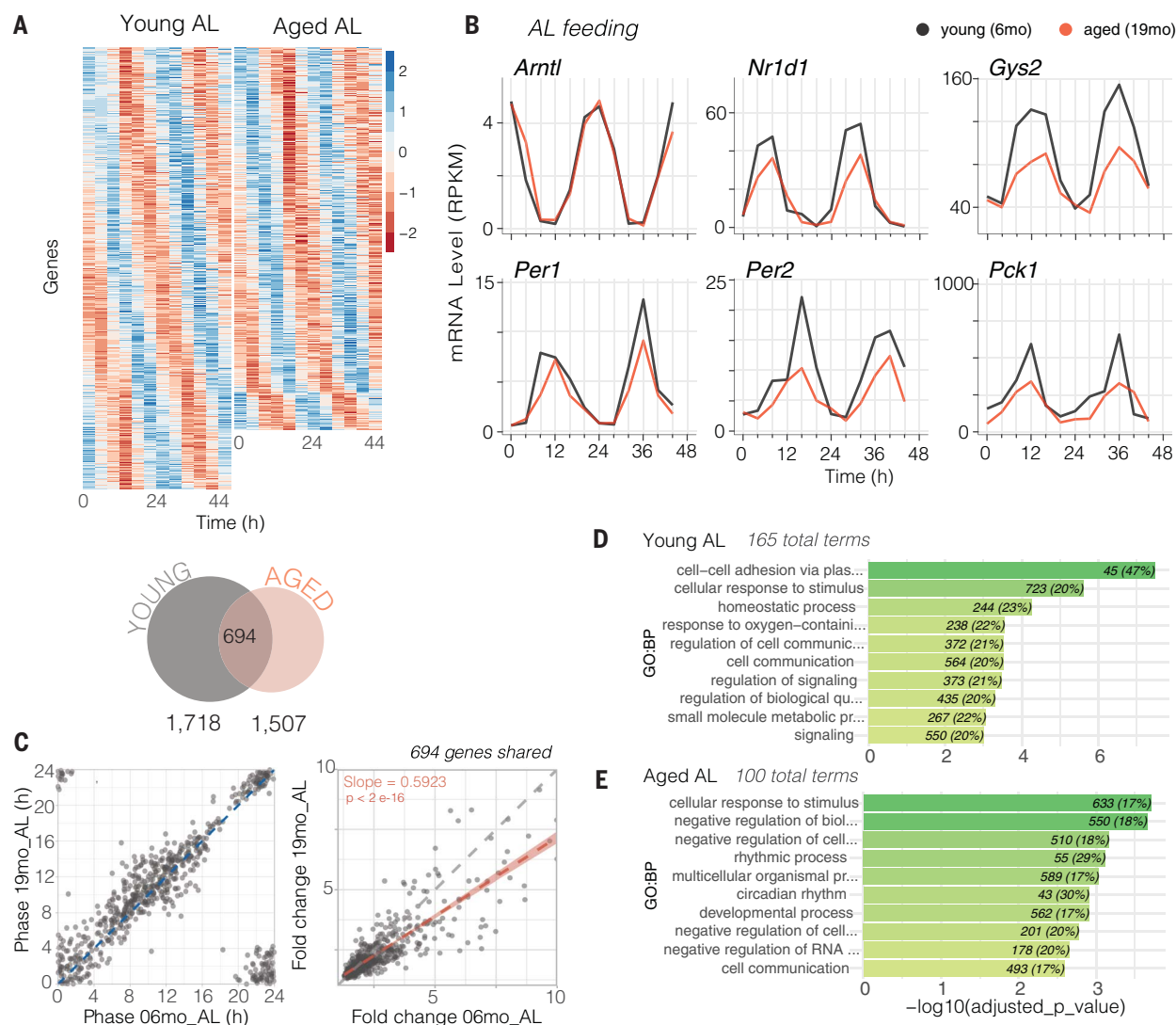


Fig. 5. Circadian rhythms in liver gene expression are blunted during aging in AL mice. (A) Gene expression patterns from mRNA-seq were analyzed for circadian rhythms using the ARSER, JTK_CYCLE (from Metacycle R Package), and RAIN circadian algorithms. Heatmaps (top) are sorted by phase of gene expression. Each row is one gene with expression level in z-score at 12 time points (columns). Venn diagram (bottom) shows the number of rhythmic genes in young (gray) and aged (red) AL livers using stringent criteria (significantly cycling according to three algorithms; Benjamini-Hochberg P and $q < 0.05$ and $\log_2FC > 0.3$) to define rhythmicity. (B) Examples of circadian profiles of genes

identified as cycling in the CR-day-2h group. Therefore, we do not recommend putting too much weight on the number of cycling genes called but rather emphasize the phase and amplitude of gene expression.

Discussion

Classic CR protocols not only reduce energy intake but also lead to severe behavioral time-restricted feeding behavior and prolonged fasting intervals (7–9). Because time-restricted feeding and fasting are beneficial to health, these two factors may contribute to lifespan extension in classic CR experiments. In the

work presented here, we deconvolved the effects of calories, fasting, and circadian alignment on longevity. We compared five different CR feeding groups that differed only in the daily pattern of food consumption without any changes in food composition or energy content. We used an automated feeding system (7) to test whether the timing and fasting period between meals affected life span in male mice under 30% CR by allowing food access only during the day, at night, or evenly distributed throughout 24 hours. By spreading out food through 24 hours, with no day-night feeding pattern in the CR-spread group, we

that are rhythmic in both young and aged AL livers. Black indicates young AL livers, and red indicates aged AL livers. **(C)** Comparison of phase (left, hours) and amplitude (right, daily fold change) of the 694 genes that were rhythmic in both age groups. The red correlation line (Spearman) and linear regression (slope is statistically different from 1; $P < 2 \times 10^{-16}$) in the fold change comparison indicates that aged animals showed overall reduced amplitude of rhythmic genes. **(D and E)** GO terms of genes that are cycling in either young (D) or aged (E) AL mice. Represented are 10 nonredundant terms of the top 25 most significant enriched terms.

found only an ~10% extension of median life span compared with AL-fed mice. The CR-day fed groups that had an ~12- or ~22-hour fasting interval lived ~20% longer than AL control mice. The degree of life-span extension was significantly longer when food was consumed during the nighttime, which is the normal feeding time in nocturnal rodents (~35% versus 20% compared with AL, log-rank Mantel-Cox $P < 0.0001$, and ~10% night versus day, log-rank Mantel-Cox $P < 0.05$) (Fig. 2A). Although potential sleep disruption needs to be carefully studied in CR-day versus CR-night groups, recent evidence shows that sleep homeostasis is

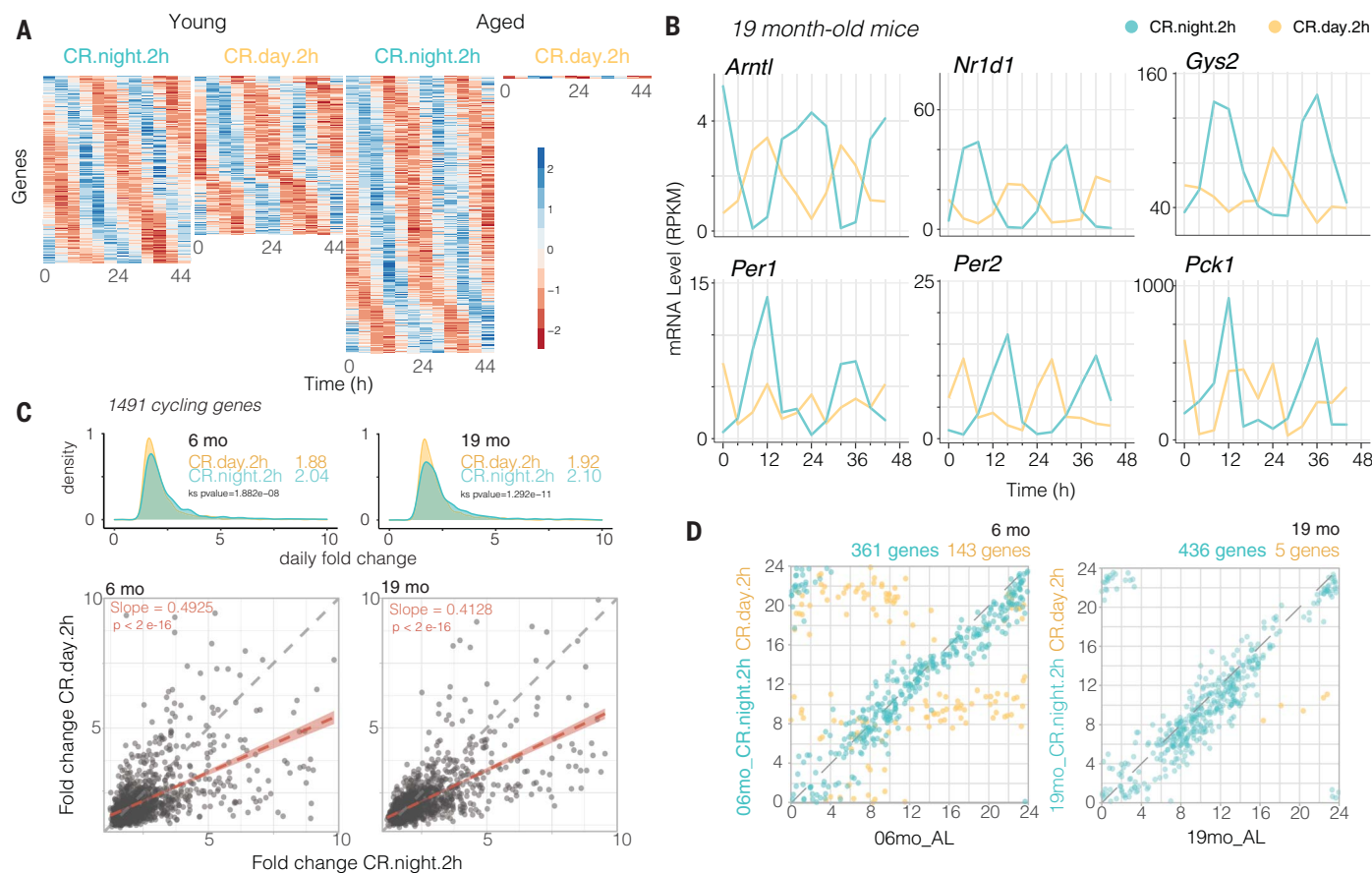


Fig. 6. Effects of CR and phase of feeding on circadian gene expression.

(A) Gene expression patterns from mRNA-seq were analyzed for circadian rhythms using the ARSER, JTK_CYCLE (from the Metacycle R Package), and RAIN circadian algorithms. Heatmaps sorted by phase of gene expression. Each row is one gene with expression level in z-score shown at 12 time points (columns). (B) Examples of the circadian profiles of the same genes shown in Fig. 5B comparing profiles from CR-night-2h (blue) to CR-day-2h (yellow) aged mice. (C) Comparison of circadian amplitude (daily fold-change) of 1491 rhythmic genes from young (left) and aged (right) CR groups. Top panels

show amplitude density plots (median amplitude values are inset). Bottom panels are correlation plots comparing amplitude of genes from CR-night-2h to CR-day-2h in young (left) and aged (right) mice. The linear regression lines have slopes that are significantly <1 ($P < 2 \times 10^{-16}$). (D) Phase correlation plots of rhythmic genes from young (left) and old (right) CR-night-2h-fed (blue) and CR-day-2h-fed (yellow) mice versus AL-fed mice of the same ages. Phase is represented in hours. Numbers of shared cycling genes between each CR condition and AL are labeled on top of the correlation plot.

maintained with daytime feeding (29). All CR groups maintained a relatively steady body weight throughout their life span, indicating that the additive effects of CR combined with appropriate timing of food consumption were independent of weight gain. Thus, the maximal pro-longevity benefits of CR can be achieved by a fasting interval of >12 hours in which a time-restricted feeding interval occurs in phase with the natural nocturnal circadian phase of feeding in mice (i.e., circadian alignment).

These results are consistent with recent studies in C57BL/6J male mice in which similar life-span extensions were reported in once-per-day CR mice fed in the morning (20%, equivalent to our CR-day-2h group) (9) or mice on $\sim 28\%$ CR when food was consumed during the daytime but closer to the normal active phase at night (9 hours later than our CR-2h-day and 3 hours earlier than our CR-2h-night groups) (8). Pak *et al.* (9) suggested that

fasting alone drives the geroprotective effects of CR, which is partially consistent with our results; however, they argued that using a 50% cellulose-diluted diet that 30% CR does not extend life span and did not test whether the phase or circadian alignment of CR could be a factor. By contrast, we showed that 30% CR without dilution of the diet in the CR-spread group extended life span by $\sim 10\%$; thus, we conclude that 30% CR alone without fasting or circadian alignment accounts for a 10% extension of life span. In our study, the diets used were identical in all six groups, which mitigates against the confounding effects of diet composition such as fiber (cellulose) and uncontrolled grain-based diets (23, 49, 50). Therefore, our conclusions differ from Pak *et al.* (9) in two ways: (i) we found that CR alone without fasting can still extend life span by $\sim 10\%$ using the same diet and (ii) we found that fasting contributes in an additive man-

ner to life-span extension rather driving the effects of CR.

In nonhuman primates, the effects of CR on longevity have differed between studies performed at the University of Wisconsin (UW) and the National Institute of Aging (NIA). However, many differences in study design, diet, and feeding protocols have been documented, and the overall conclusions from a joint consensus is that CR improves health and survival in rhesus monkeys (51–53). CR did not extend life span in the NIA study; however, the AL control group was longer lived, slightly caloric restricted, food was provided twice a day, and body weights were lower than AL controls in the UW study. In addition, in the UW study, food was provided AL during the day from $\sim 8:00$ a.m. to $\sim 4:00$ p.m., but was removed during the night. Thus, in both studies, the AL groups were either partially calorie restricted (NIA) or

were under a time-restricted feeding protocol (UW), which should be beneficial. Thus, it would be of considerable interest to explore the effects of time-restricted feeding and circadian phase of CR in future studies in non-human primates.

The benefits of circadian alignment of feeding and fasting cycles are widespread across species. It increases life span in flies (54), protects against metabolic disorders in mice (17, 18, 55, 56), and promotes health in humans (57, 58). We also found that CR promotes widespread metabolic benefits, which include lower body weight with reduced fat content. Furthermore, CR (particularly the longest-lived group, CR-night) attenuated age-related changes observed under AL by improving glucose homeostasis, insulin sensitivity, and hormonal profiles.

Using circadian gene profiling, we found that timing of food intake led to complex genome-wide reprogramming of circadian gene expression in the liver, which is consistent with other findings (11, 14, 33, 59–61). This emphasizes the importance of considering the sampling time, which is often a snapshot, before concluding whether any intervention increases or decreases the expression of a gene of interest. If a gene has a circadian oscillation, then a single snapshot could lead to the opposite conclusions depending on the time it was taken, particularly in classic CR protocols resembling our CR-day-2h group (1, 7).

Aging was associated with increased expression of the genes involved in immune processes and inflammation and decreased expression of the genes involved in metabolism and circadian biology. CR treatment restored or attenuated many of these age-related changes in gene expression in a manner similar to results reported by the de Cabo (24, 34) and Sassone-Corsi (14) groups. Decreases in circadian amplitude of gene expression with age were reduced by CR. Overall, feeding time and fasting had additive effects on CR-mediated life-span extension. Together, CR and time restriction of feeding to the nighttime optimally extended life span and delayed many of the age-related gene expression changes in immune function, inflammation, and metabolism. Further studies are required to determine whether disruption of circadian sleep/wake cycles by daytime feeding also contributes to the day versus night differential responses in the liver transcriptome and life span. Thus, circadian interventions such as timed feeding can enhance the well-known life-span benefits of CR.

There are many links between the circadian system and aging (20). High-amplitude circadian rhythms correlate with well-being (13, 62, 63), whereas clock dysfunction leads to metabolic disorders, premature aging, and reduced life span (42, 64–67). During normal

aging, rhythms decrease in amplitude and often exhibit a shift in phase (14, 20, 68–72). In both flies and mice, the effects of dietary restriction on life span require core circadian clock genes (73–75). The circadian system regulates most of the metabolic pathways implicated in longevity (3, 4, 13, 22, 32, 59, 76, 77). Molecules known to function in the regulation of life span by dietary restriction, such as insulin and insulin-like growth factor 1 (IGF-1), Sirtuin1 (SIRT1), nicotinamide phosphoribosyltransferase (NAMPT), AMP-activated protein kinase (AMPK), PPAR γ coactivator 1 (PGC-1 α), mechanistic target of rapamycin (mTOR), and glycogen synthase kinase 3 β (GSK3 β), are all intricately involved in the molecular mechanisms of circadian clocks (77–89). The master circadian transcription factors CLOCK and BMAL1 have direct gene targets in every fundamental metabolic pathway in the liver (25, 32, 90, 91). Because of these direct links among the pathways involved in aging and longevity, metabolism, and the circadian clock, our results demonstrate the importance of timing of CR and indicate that optimizing the phase of circadian gene expression may be a powerful intervention for extending life span.

We used C57BL/6J male mice in this study. However, there could be strain- and sex-specific responses worth studying further (24, 92), because, for example, ovarian hormones can protect females against dietary challenges that otherwise disrupt circadian rhythms in males (55). In future work, both sexes and multiple genetic backgrounds (92–94) should be used to explore the broader effects of circadian interventions on aging, and the results may support the application of circadian-timed interventions in human studies.

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Competing interests: The authors declare no competing interests. **Data and materials availability:** Data are available in the main text or the supplementary materials. RNA-seq data

have been deposited in the Gene Expression Omnibus (accession no. GSE190939). **License information:** Copyright © 2022 the authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original US government works. <https://www.science.org/about/science-licenses-journal-article-reuse>

SUPPLEMENTARY MATERIALS

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Materials and Methods

Figs. S1 to S14

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Data S1 to S6

MDAR Reproducibility Checklist

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Circadian alignment of early onset caloric restriction promotes longevity in male C57BL/6J mice

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Timing eating and fasting for longevity

Animals fed a limited number of calories, just enough to avoid malnutrition, show extended health span and life span. However, they are so hungry that they eat those fewer calories in a limited period of time and consequently spend more time fasting than do animals for which access to food is not restricted. Acosta-Rodríguez *et al.* therefore designed experiments in mice to control both caloric intake and the timing of their eating to see which factors were the most important (see the Perspective by Deota and Panda). Caloric restriction extended life span as expected, but it worked best when feeding was restricted so that the animals fasted for at least 12 hours and when the period in which the animals ate corresponded to the active phase of their circadian cycle. —LBR

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